CHAMHEMBE

CAR RENTAL

Fleet & Rental Management System

Technical Proposal & System Blueprint

Prepared by Leading Digital

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Version 1.0

1. Executive Summary

Chamhembe Car Rental requires a purpose-built digital platform to manage its mixed fleet model **(owned + outsourced vehicles)**, streamline bookings, enforce mileage-based pricing, track compliance documents (ZINARA, ZBC, insurance, fitness), and record all maintenance and breakdown costs against individual vehicles. The system must serve both internal operations (admin dashboard) and customers (web booking portal + mobile app).

This proposal outlines a comprehensive Fleet & Rental Management System built on Laravel 12 (API + admin), React (web dashboard + customer portal), and React Native (iOS/Android mobile app) — your proven production stack.

2. Business Context & Problem Statement

Chamhembe operates a hybrid model: a core fleet of owned vehicles supplemented by outsourced vehicles from third-party owners. For outsourced vehicles, Chamhembe either applies a markup or retains the same price to secure the client relationship. This creates several operational challenges:

- No centralised view of fleet availability across owned and outsourced vehicles
- Manual mileage tracking makes it difficult to calculate overage charges against the 200 km/day included allowance
- Compliance documents (ZINARA tolls, ZBC radio licence, vehicle fitness, insurance) are tracked ad-hoc with risk of expiry
- Breakdown and maintenance costs are not tied to individual vehicle P&L records
- Customer accounts, booking history, and payment records are fragmented

The proposed system eliminates these gaps with a single platform covering fleet operations, customer management, booking, compliance, and financials.

3. System Overview

The platform consists of three application layers sharing a single Laravel API backend:

Layer	Technology	Users
Admin Dashboard	React + Inertia.js	Chamhembe staff: fleet managers, finance, operations
Customer Portal	React (SPA)	Walk-in and online customers browsing availability and booking
Mobile App	React Native	Customers on iOS/Android for bookings, trip tracking, payments

4. Core Modules

4.1 Fleet Management

Centralised registry of every vehicle — owned or outsourced — with full lifecycle tracking.

Feature	Description
Vehicle Registry	Add/edit vehicles with make, model, year, colour, registration plate, VIN, fuel type, transmission, seating capacity, category (sedan/SUV/truck/van), and photos.
Ownership Type	Flag each vehicle as Owned or Outsourced. For outsourced: record vehicle owner contact, agreed rate, and Chamhembe markup percentage or fixed margin.
Vehicle Status	Real-time status: Available, Rented, In Maintenance, Reserved, Decommissioned. Drives availability calendar.
Vehicle Timeline	Full audit log per vehicle: every rental, service, breakdown, document upload, and status change with timestamps.
GPS Integration	Optional CarTrack or similar GPS integration for live location and trip distance logging via API.

4.2 Booking & Reservations

End-to-end booking flow from search to return, with calendar-driven availability.

Feature	Description
Availability Calendar	Visual calendar showing each vehicle's bookings, maintenance blocks, and open slots. Filterable by category, dates, and price range.
Online Booking	Customers select dates, vehicle category, and optional extras (GPS tracker, child seat, cross-border permit). System calculates cost based on daily rate + included 200 km/day.
Reservation Workflow	Pending → Confirmed (on deposit) → Active (vehicle collected) → Completed (vehicle returned) → Invoiced. Each transition triggers notifications.
Walk-in Bookings	Admin can create bookings on behalf of walk-in customers directly from the dashboard.
Rental Agreement	Auto-generated PDF rental agreement with customer details, vehicle info, included mileage, excess rate, insurance terms, and digital signature capture.
Calendar Sync	Export bookings to Google Calendar / iCal for managers.

4.3 Mileage & Pricing Engine

Automated mileage tracking and overage calculation against the 200 km/day included allowance.

Feature	Description
Odometer Logging	Record odometer reading at vehicle pickup and return. System calculates total trip distance automatically.

Included Allowance	Default 200 km/day (configurable per vehicle or booking). System computes total allowance as rental days × daily limit.
Overage Calculation	If actual km > allowance, excess km × per-km rate is added to the invoice. Rate configurable globally or per vehicle category.
Trip Log	Detailed log of each trip segment if GPS is active, or manual entry of intermediate readings for multi-day rentals.
Pricing Rules	Flexible pricing: daily, weekly, monthly rates. Weekend/holiday surcharges. Long-term discount tiers. Outsourced vehicle markup applied automatically.
Fuel Policy	Record fuel level at pickup/return. Charge for missing fuel at a configurable per-litre rate.

4.4 Licensing & Compliance

Simple record-keeping for every regulatory licence, certificate, and insurance policy per vehicle. No document uploads — just the key data: what it is, what it cost, and when it expires.

Feature	Description
ZINARA Licence	Record licence number, issue date, expiry date, and cost paid. Alert 30/14/7 days before expiry.
ZBC Radio Licence	Record licence number, issue date, expiry date, and cost paid. Same tiered alert schedule.
Vehicle Fitness	Certificate number, test date, expiry date, cost paid, and testing station.
Insurance	Provider (e.g., Old Mutual, Zimnat, Nicoz Diamond), policy number, type (comprehensive/third-party), cover amount, premium cost, and expiry date. Multi-policy support per vehicle.
Vehicle Registration	Registration plate, registration date, renewal date, and cost.
Custom Licence Types	Add any other licence or permit type as needed (e.g., cross-border permit, defensive driving cert). Flexible type system so you're not locked to a fixed list.
Cost Tracking	Every licence/renewal records what was paid and when, feeding into the per-vehicle cost ledger for profitability calculations.
Compliance Dashboard	Single view of all vehicles with colour-coded expiry status: green (valid >30 days), amber (expiring within 30 days), red (expired). Bulk export to CSV.
Alerts & Notifications	Email + in-app + push notification alerts for upcoming and overdue items. Escalation to manager if not actioned within the alert window.

4.5 Maintenance, Breakdowns & Operational Costs

Every cost incurred on a vehicle is captured here — from scheduled servicing to breakdowns, car washes, towing, accident damage, and any ad-hoc expense. Everything ties back to the vehicle's P&L.

Feature	Description

Scheduled Service	Set service intervals by mileage (e.g., every 10,000 km) or time (e.g., every 6 months). System triggers alerts based on odometer readings and dates.
Service Records	Log each service: date, odometer, service type (oil change, brake pads, tyres, etc.), service provider, cost, and notes.
Breakdown Reporting	Record breakdowns with: date, location, description, repair cost, parts replaced, downtime days, and whether customer was relocated to another vehicle.
Towing Costs	Dedicated towing cost field on breakdowns and accidents. Record tow company, distance, cost, and link to the incident.
Accident Records	Full accident log: date, location, description, third-party details, police report reference, damage assessment, repair cost, insurance claim status, and customer liability.
Car Wash	Log car wash expenses per vehicle: date, wash type (basic/full/valet), cost, and provider. Track frequency and spend over time.
Other / Ad-hoc Costs	Flexible expense entry for anything else: parking fines, toll fees, replacement keys, interior cleaning, dent removal, windscreen repair, etc. Custom cost categories that you can add as your business grows.
Cost Categories	Pre-defined and custom categories for all expenses: Maintenance, Breakdown, Towing, Accident, Car Wash, Fuel Top-up, Fines, Licensing, Other. Used for filtering, reporting, and vehicle P&L breakdowns.
Vendor Management	Directory of service providers (mechanics, tow companies, car wash outlets, parts suppliers) with contact details and cost history.
Vehicle Downtime	Automatically blocks vehicle availability during maintenance/repair periods. Calculates total downtime days per vehicle for fleet efficiency metrics.

4.6 Customer Management

Every customer gets an account with full booking history, documents, and financial records.

Feature	Description
Customer Accounts	Name, ID number, phone, email, physical address, driver's licence details (number, class, expiry), emergency contact.
KYC Documents	Upload and store: national ID, driver's licence, proof of residence. Required before first booking.
Booking History	Full history of all past and current rentals with vehicle, dates, mileage, cost, and payment status.
Customer Wallet	Prepaid balance / deposit system. Customers can top up and book against balance. Refunds credited to wallet.
Blacklist	Flag problematic customers (e.g., vehicle damage, non-payment) with notes. Blocked from future bookings.
Corporate Accounts	Support for business accounts with negotiated rates, monthly invoicing, and multiple authorised drivers.

Loyalty Programme	Points-based rewards: earn points per rental day, redeem for free days or upgrades.

4.7 Financial Module

Invoicing, payments, and profitability tracking across the fleet.

Feature	Description
Invoicing	Auto-generated invoices on booking completion. Line items: base rental, mileage overage, fuel charges, extras, damage charges. PDF export.
Payment Tracking	Record deposits, partial payments, and final settlements. Support for cash, EcoCash, bank transfer, and card (via Paynow Zimbabwe).
Outsource Payouts	For outsourced vehicles: calculate owner payout (agreed rate) and Chamhembe commission per booking. Track payable/paid status.
Vehicle P&L	Per-vehicle profitability: total revenue vs. total costs (maintenance, insurance, breakdowns, fuel). Monthly and lifetime views.
Revenue Dashboard	Fleet-wide revenue, occupancy rate, average daily rate (ADR), revenue per available vehicle (RevPAV).
Tax Reports	Summary reports for ZIMRA tax compliance. Revenue by period, expense categories, and VAT calculations.

4.8 Notifications & Communication

Multi-channel notifications to keep staff and customers informed at every stage.

Feature	Description
Booking Confirmations	Email + SMS + push notification on booking confirmation, reminder 24 hours before pickup.
Document Expiry Alerts	Tiered alerts to admin for ZINARA, ZBC, insurance, and fitness expiry.
Service Due Alerts	Notification when a vehicle approaches its next scheduled service interval.
Payment Reminders	Automated reminders for outstanding balances and overdue invoices.
WhatsApp Integration	Optional WhatsApp Business API integration for booking confirmations and customer communication (popular in Zimbabwe).

5. Technical Architecture

5.1 Stack Summary

Component	Technology

Backend API	Laravel 12 (PHP 8.3) — RESTful API with Laravel Sanctum authentication
Admin Dashboard	React + Inertia.js (server-driven SPA, consistent with your existing stack)
Customer Web Portal	React SPA consuming the Laravel API
Mobile App	React Native (iOS + Android) with shared component library
Database	MySQL 8 (primary), Redis for caching, queues, and real-time pub/sub
Real-time	Laravel Reverb for WebSocket-driven live availability updates and notifications
File Storage	Local disk or S3-compatible (e.g., Backblaze B2) for documents, photos, receipts
Payments	Paynow Zimbabwe (EcoCash, OneMoney, card) + manual capture for cash/bank transfer
SMS Gateway	Twilio or local provider (e.g., Telerivet, Bongolive) for transactional SMS
Hosting	DigitalOcean Droplet (Ubuntu) with Cloudflare CDN, or cPanel/WHM if preferred
CI/CD	GitHub Actions → automated tests → deploy to DigitalOcean
GPS (Optional)	CarTrack API for live vehicle location and automated mileage logging

5.2 Database Design (Key Entities)

The following core tables anchor the data model. All tables include standard Laravel timestamps and soft deletes.

Entity	Key Fields
vehicles	make, model, year, reg_plate, vin, category, fuel_type, transmission, seats, ownership_type (owned/outsourced), owner_id (nullable), daily_rate, weekly_rate, monthly_rate, km_per_day_limit, excess_km_rate, status, current_odometer, photos
vehicle_owners	name, phone, email, bank_details, agreed_daily_rate, commission_type (percentage/fixed), commission_value
customers	name, id_number, phone, email, address, licence_number, licence_class, licence_expiry, wallet_balance, status (active/blacklisted), corporate_account_id
bookings	customer_id, vehicle_id, pickup_datetime, return_datetime, odometer_start, odometer_end, km_allowance, status (pending/confirmed/active/completed/cancelled), total_amount, notes
invoices	booking_id, customer_id, subtotal, mileage_overage, fuel_charge, extras, damage_charge, tax, total, status (draft/sent/paid/overdue)
payments	invoice_id, amount, method (cash/ecocash/bank/card), reference, paynow_poll_url, status, paid_at
vehicle_licences	vehicle_id, type (zinara/zbc/fitness/insurance/registration/custom), label, document_number, provider, issue_date, expiry_date, cost, notes, status

vehicle_costs	vehicle_id, booking_id (nullable), category (maintenance/breakdown/towing/accident/car_wash/fuel/fine/licensing/other), description, amount, vendor_id (nullable), odometer, incident_date, notes
maintenance_records	vehicle_id, type (scheduled/breakdown/accident), description, odometer, service_provider, repair_cost, parts_cost, tow_cost, downtime_days, insurance_claim_ref, completed_at
mileage_logs	vehicle_id, booking_id, odometer_reading, recorded_at, source (manual/gps), notes

5.3 API Structure

The API follows RESTful conventions with versioned endpoints under /api/v1/. Key resource groups:

- /api/v1/auth — Registration, login, password reset (Sanctum token-based)
- /api/v1/vehicles — CRUD, availability check, status updates, document management
- /api/v1/bookings — Create, confirm, cancel, extend, complete, with availability validation
- /api/v1/customers — Profile, KYC documents, wallet, booking history
- /api/v1/invoices — Generate, send, payment recording
- /api/v1/payments — Paynow initiation, webhook handling, manual capture
- /api/v1/maintenance — Service records, breakdown reports, cost logging
- /api/v1/reports — Fleet utilisation, revenue, compliance, vehicle P&L
- /api/v1/notifications — User notification preferences and history

6. User Roles & Permissions

Role-based access control (RBAC) using Laravel's built-in policies and gates, with Spatie Laravel Permission for role management:

Role	Access	Typical User
Super Admin	Full system access, settings, user management, financial reports	Kudzai / business owner
Fleet Manager	Vehicle CRUD, maintenance, compliance docs, availability management	Operations staff
Booking Agent	Create/manage bookings, customer accounts, walk-in rentals	Front desk / reception
Finance	Invoices, payments, payouts, revenue reports, tax exports	Accounts / bookkeeper
Customer	Browse vehicles, book, view history, make payments, rate experience	Rental customers

7. Mobile App (React Native)

The mobile app targets customers and provides a streamlined booking and trip management experience:

- Vehicle browsing with filters (category, price, dates, availability)
- Booking flow: select vehicle → choose dates → review pricing → pay deposit → confirmation
- Digital rental agreement with signature capture
- Trip dashboard: current booking, vehicle details, odometer entry, mileage tracker
- Payment via EcoCash, OneMoney, or card through Paynow SDK
- Push notifications for booking confirmations, reminders, payment receipts
- Booking history and invoice downloads
- In-app support chat or WhatsApp deeplink
- Rate and review experience after each rental

Admin Companion (Phase 2): A simplified admin mobile app for fleet managers to update vehicle status, log odometer readings, and approve bookings on the go.

8. Mileage Pricing Worked Example

To illustrate the mileage overage engine, here is a sample calculation:

Parameter	Value
Vehicle	Toyota Fortuner (SUV)
Daily Rate	USD $80
Rental Period	5 days
Included Mileage	200 km/day × 5 days = 1,000 km
Odometer at Pickup	45,230 km
Odometer at Return	46,450 km
Actual Distance	1,220 km
Excess Mileage	1,220 – 1,000 = 220 km
Excess Rate	USD $0.35 per km
Mileage Overage Charge	220 × $0.35 = USD $77.00
Base Rental	5 × $80 = USD $400.00
Total Invoice	**USD $400.00 + $77.00 = USD $477.00**

9. Phased Delivery Plan

Recommended phased approach to get core functionality live quickly, then iterate:

Phase 1: Foundation (Weeks 1–6)

- Laravel API scaffolding with authentication (Sanctum)
- Vehicle registry and fleet management CRUD
- Customer accounts and KYC document uploads
- Basic booking flow with availability calendar
- Admin dashboard (React + Inertia.js) for fleet and booking management
- Database migrations, seeders, and test data

Phase 2: Operations (Weeks 7–10)

- Mileage tracking and overage calculation engine
- Compliance document management with expiry alerts
- Maintenance and breakdown recording
- Invoicing with PDF generation
- Paynow Zimbabwe payment integration (EcoCash, card)

Phase 3: Customer-Facing (Weeks 11–14)

- Customer web portal (React SPA) with booking and payment
- React Native mobile app (customer-facing)
- Push notifications and email/SMS alerts
- Digital rental agreement with signature capture
- Booking reminders and payment receipts

Phase 4: Intelligence & Growth (Weeks 15–18)

- Vehicle P&L and fleet profitability reports
- Revenue dashboard with occupancy and ADR metrics
- Outsourced vehicle payout management
- Loyalty programme and corporate accounts
- WhatsApp Business API integration
- GPS/CarTrack integration for automated mileage
- Admin mobile companion app

10. Zimbabwe-Specific Considerations

The system is tailored for the Zimbabwean market:

- **Multi-currency:** Support for USD, ZiG, and bond notes. Configurable default currency per booking. Exchange rate management for cross-currency invoicing.
- **ZINARA Integration:** Toll fees are a significant operating cost. Track ZINARA licence per vehicle and optionally log toll expenses against trips.
- **ZBC Compliance:** Radio licence tracking and renewal reminders to avoid fines.

- **Insurance Providers:** Pre-loaded list of Zimbabwean insurers (Old Mutual, Zimnat, Nicoz Diamond, FBC Insurance, CABS) for quick selection.
- **Payment Methods:** EcoCash and OneMoney are dominant mobile money platforms. Paynow Zimbabwe is the aggregator for card and mobile money payments.
- **Cross-border Rentals:** Flag bookings going outside Zimbabwe (e.g., to South Africa, Mozambique, Zambia). Require additional insurance and border documentation.
- **Power & Connectivity:** Mobile app designed for intermittent connectivity with offline-capable booking drafts and queued sync.

11. Security & Data Protection

- Laravel Sanctum for API token authentication with ability-based scoping
- Role-based access control via Spatie Laravel Permission
- All customer PII and financial data encrypted at rest (AES-256)
- HTTPS enforced across all endpoints (Cloudflare SSL)
- File uploads scanned and stored outside web root with signed temporary URLs
- Rate limiting on API endpoints to prevent abuse
- Audit trail on all sensitive operations (who did what, when)
- Regular automated backups to off-site storage
- POPIA/data protection principles applied (consent, minimisation, retention limits)

12. Next Steps

To move forward with the Chamhembe Car Rental system, the following actions are recommended:

1. **Review & Feedback:** Review this proposal and confirm module priorities, any features to add/defer, and budget parameters.
2. **Branding & Design:** Confirm Chamhembe brand colours, logo, and design direction for the customer-facing portal and mobile app.
3. **Fleet Data:** Provide a list of current vehicles (owned + outsourced) with rates, so we can seed the system with real data from day one.
4. **Payment Gateway:** Set up a Paynow Zimbabwe merchant account if not already in place.
5. **Kick-off:** Schedule a kick-off session to finalise scope, timeline, and begin Phase 1 development.

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